

Research Methods

Summary

AEPSY – 2019

Yr 11 Psychology

*Please refer to your flashcards to enhance your memory of key words and content

* Refer to this booklet for examination/test type questions and further notes on Research Methods

Research Methods Booklet AEPSY



From your syllabus:

Planning and conducting psychological research

- Research terminology
 - experimental, non-experimental
 - scientific, non-scientific
 - sample
 - population
- Ethics in psychology research
 - informed consent
 - confidentiality
 - voluntary participation and withdrawal rights
 - deception in research
- Steps in the scientific method
- Differences between sample and population data
- Experimental research methods
 - independent and dependent variables
 - operational hypotheses
 - controlled and uncontrolled variables
 - experimental and control groups
 - reliability and validity

- Non-experimental (descriptive) research methods
 - case studies, surveys, correlational studies and archival research
 - behavioural variables (not dependent and independent variables) in correlational studies
- Qualitative methods of data collection
- Quantitative methods of data collection – fixed response, rating scales

Processing and evaluating psychological research

- Methods of displaying quantitative data – tables, graphs and diagrams
- Interpretation of the following forms of data:
 - mode
 - mean
 - median
 - range
- Role of probability
- Use of correlation to establish association between variables
- Sources of error in data and ways of reducing them
- Evidence-based conclusions related to the hypothesis

RESEARCH TERMINOLOGY (Chapters 10 and 11)

Experimental research: research that allows a researcher to manipulate (change) the independent variable. A cause-and-effect relationship can be found. (Cause and effect relationship refers to changes in an independent variable causing differences in a dependent variable). Experimental research allows for the random assignment of participants to conditions.

Non-experimental research: research that does not allow the researcher to manipulate the independent variable. A cause-and-effect relationship cannot be found and instead researchers may rely on interpretation, observation or interactions to come to a conclusion. Non-experimental research does not allow for the random assignment of participants to conditions.

See Page 38 of this booklet for more information on experimental vs non experimental classifications.

Scientific research: research that follows scientific method and the findings of scientific research can be reproduced and demonstrated to be consistent (the results can be reliable).

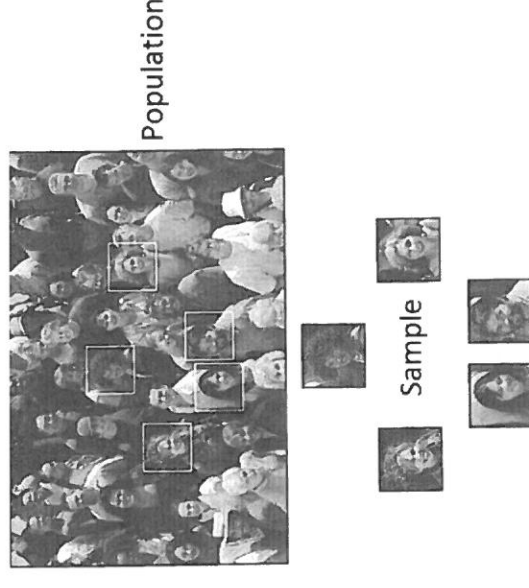
Non-scientific research: research that does not follow scientific method and cannot be reproduced and demonstrated to be consistent (the results are not reliable).

Population: the complete collection of people, objects or events that can possibly be measured. The population is the entire group of people belonging to a particular category (for example, all university students or all AFL footballers). It is the larger group of research interest from which a sample is to be drawn.

• In this context, the term *population* does not refer to the number of people living in a particular area (such as the population of Western Australia), but rather a group of people with similar characteristics that are of interest to the researcher.

• For example, if you were to conduct a study investigating Western Australian year 12 students' favourite subjects, the population would be all the students who are enrolled in year 12 in schools in Western Australia.

Sample: a group of participants selected from, and representative of, a population of research interest. A sample must represent the population from which it is drawn in order for inferences to be made about that population. The process of choosing participants from the population for use in a study is called participant selection, or sampling. A sample is a subsection of a population, and therefore it is a smaller



group than the population itself.

Difference between sample and population data

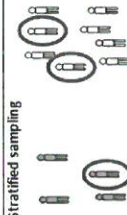
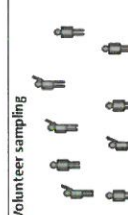
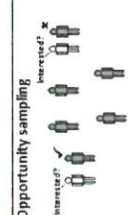
Data: information (observable facts) that psychologists systematically collect in studies, investigations and experiments.

Population data: data collected from a whole population that is defined by the researcher.

- ♦ For example, the height for all women in the U.S ages 18-35 or the weekly time spent online in hours by all children ages 11-17.
- ♦ A large population may be impractical to study because of the time it would take and the high cost.

Sample data: data collected from the sample that is being studied.

- ♦ A small but well-chosen sample can accurately reflect the characteristics of the entire population from which it is chosen.

Random sampling 	Every member of a population has an equal chance of being selected E.g. Pulling names out of a hat	For very large samples it provides the best chance of an unbiased representative sample	For large populations it is time-consuming to create a list of every individual.
Stratified sampling 	Dividing the target population into important subcategories Selecting members in proportion that they occur in the population E.g. 2.5% of Britain are of Indian origin, so 2.5% of your sample should be of Indian origin... and so on	A deliberate effort is made to make the sample representative of the target population	It can be time consuming as the subcategories have to be identified and proportions calculated
Volunteer sampling 	Individuals who have chosen to be involved in a study. Also called self-selecting E.g. people who responded to an advert for participants	Relatively convenient and ethical if it leads to informed consent	Unrepresentative as it leads to bias on the part of the participant. E.g. daytime TV advert would not attract full-time workers.
Opportunity sampling 	Simply selecting those people that are available at the time. E.g. going up to people in cafés and asking them to be interviewed	Quick, convenient and economical. A most common type of sampling in practice	Very unrepresentative samples are often biased by the researcher who will likely choose people who are "helpful"

Collecting sample data is more manageable and less expensive to study.

Sampling methods

- ♦ **Convenience sampling:** is a method of sampling in which participants are chosen because they are close by and easily accessed.
- ♦ **Random sampling:** is a common sampling method that ensures that each member of the population of research interest has an equal chance of being selected into a study. This method is to ensure that the sample is representative of the population. This can be done through using a series of random numbers (manually selected like Lotto, from a random numbers table or from a computer-generated program) to determine group allocation, by tossing a coin, or by drawing names out of a hat.

♦ Stratified sampling: is a method of sampling used when there are several groups or 'strata' in a population and it is important to represent these groups in the same proportions in the sample.

♦ Random-stratified sampling: the population is divided into a number of strata according to some characteristic of interest related to the variable(s) being studied (for example, age, gender or income). Then a list of all persons within each stratum is obtained. Simple random samples are then selected from each stratum to ensure that each member of the group has an equal chance of being selected for the sample. Random samples of proportionate size are drawn from within each stratum, making the sample a proportionately-stratified random sample.

Generalisation

- ▶ Having a sample that represents the population is important because otherwise the results will not generalize well beyond the sample.

ETHICS

Informed Consent

- ▶ Informed consent provides participants with sufficiently detailed information on the study so that they can make an informed, voluntary and rational decision to participate. This includes:
 - ✓ The purpose of the study
 - ✓ Expected duration
 - ✓ Procedures of the study
 - ✓ Information on their right to decline or withdraw
 - ✓ Foreseeable consequences of withdrawing or declining
 - ✓ Potential risk, discomfort or adverse effects
 - ✓ Prospective research benefits
 - ✓ Incentives, such as payment or rewards
 - ✓ Whom to contact for questions
- ▶ Aspects must be provided to participants before they are entered into the study.

- ▶ Most researchers use a written form that the participants sign and date because there might be problems later on.
- ▶ **Special Considerations**
 - ▶ **Parental consent** – for children under the age of 18
 - ▶ **Deception** means intentionally misleading, providing incorrect information or omitting relevant information. A researcher who uses deception must still ensure that no harm comes to the subject and the act of misleading or omitting must not cause undue suffering.
- ▶ This always requires full disclosure following the experiment in the form of a **debriefing**.
 - ▶ This procedure is conducted after the experiment or study has been concluded.
 - ▶ Debriefing involves a *structured* or *semi-structured* verbal conversation between the researcher and the subjects whereby an array of topics are addressed and discussed, and the subjects are given an opportunity to ask questions.

Maintaining Privacy, Anonymity & Confidentiality

- ▶ The protection of personal information provided to the psychologist that cannot be released unless established criteria are met.
- ▶ **Researcher has access to the minimum amount of information necessary to conduct the research**
- ▶ **The methods used to obtain information about participants.**

Confidentiality

- ▶ Protect confidential information obtained through or stored in any medium
- ▶ Anything signed or noted on by the participants must be kept locked and secured.
- ▶ This often means retained files are kept in a locked filing cabinet or a password protected file on a separate drive for up to seven years after the study is concluded.
- ▶ How the data will be used and who has access to it

Anonymity

- ▶ Similar to privacy and is defined as the protection of an individual subject's identity.
- ▶ People will often be more truthful when they don't think they will be judged by how they respond, and that is what researchers really want to study.

Withdrawal from an Investigation

- ▶ Participants should be able to leave a study at any time if they feel uncomfortable.
- ▶ They should also be allowed to withdraw their data.
- ▶ They should be told at the start of the study that they have the right to withdraw.
- ▶ They should not have pressure placed upon them to continue if they do not want to

Steps in Scientific Research

1. Identify a RESEARCH issue or problem to investigate.
2. Develop the exact research question or the HYPOTHESIS
3. Choose a research DESIGN and method.
4. Collect DATA.
5. Examine the EVIDENCE.
6. ANALYSE the results.
7. COMMUNICATE the findings.

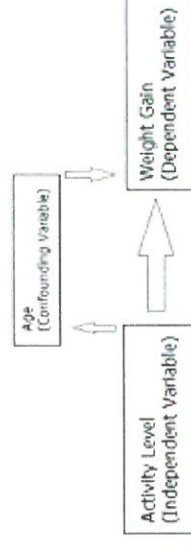
Features of Experimental Research Methods

- ♦ A variable is any condition that can change.
 - ♦ The independent variable: a condition that the experimenter manipulates (changes) in order to gauge its effect on another variable.
 - ♦ An independent variable is a suspected cause of differences in behaviour or results between groups of participants in an experiment.
 - ♦ The dependent variable: is the condition in an experiment that is affected by the independent variable.
 - ♦ The dependent variable is used to measure the independent variable's effect. A dependent variable 'depends' on the independent variable, to determine whether it changes and by how much.
- Identify the independent and dependent variables in the following list of hypotheses.*

- ♦ An extraneous variable: is a condition in an experiment that is not controllable. For example the weather, a person's diet or physical health.

- ♦ An extraneous variable: can have a hidden effect on an experiment's outcome.

Let's say you test 200 volunteers (100 men and 100 women). You find that lack of exercise leads to



weight gain. One problem with your experiment is that it lacks any control variables. For example, the use of placebos, or random assignment to groups. So you really can't say for sure whether lack of exercise leads to weight gain. One confounding variable is how much people eat. It's also possible that men eat more than women; this could also make sex a confounding variable. Nothing was mentioned about starting weight, occupation or age either. A poor study design like this could lead to bias. For example, if all of the women in the study were

middle-aged, and all of the men were aged 16, age would have a direct effect on weight gain. That makes age a confounding variable.

Controlled and uncontrolled variables

- ♦ **Controlled variable:** is a variable whose influence has been eliminated from an experiment so that it cannot affect results.
- ♦ **Uncontrolled variable:** a randomly occurring variable that might cause changes in the dependent variable.
- ♦ An uncontrolled variable is a variable whose influence has not been eliminated from an experiment.
- ♦ Uncontrolled variables may include differences between research participants, such as memory, motivation or personality. They can also include differences in how the experimenter treats the participants or how the participants react to the experimental environment. Another name for uncontrolled variable is **confounding variable**.
- ♦ Uncontrolled variables may be present without the researcher knowing, and they may not be identified (if at all) until after the experiment is complete.

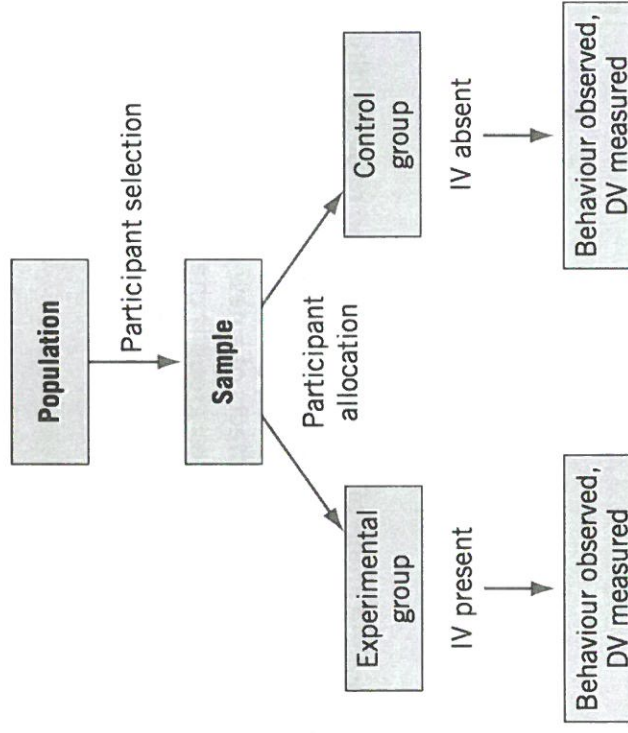
Experimental and Control groups

- ♦ **Experimental group:** the group exposed to the independent variable.
- ♦ **Control group:** the group of participants exposed to all conditions or variables except the independent variable.
- ♦ For example, in a test of the effects of a new drug, the experimental group is the group given that drug. The control group is the group that is exposed to the control condition; that is, where the variable under investigation is absent. The control group is often given no treatment, or is given a sugar pill or fake treatment rather than the new drug (placebo).

- ♦ The control group allows you to see if the drug (independent variable) has had effect on the experimental group's behaviour (dependent variable).

Why it is important to have a control group

- ♦ It is important to have a control group because: it provides a standard that the performance of the experimental group can be compared with in order to determine if the independent variable had an effect on the dependent variable.
- ♦ Without the control group, an experimenter would have no idea if the independent variable had an effect on the dependent variable or if the change would have occurred naturally or due to other factors altogether.



♦ First of all, remember that a hypothesis needs to include both the independent and dependent variable.

- ♦ The independent variable is the variable that is being changed.
- ♦ The dependent variable is the variable that is being measured.

♦ Operationalisation of a variable means that it is stated in terms that show how it is measured. For example:

Age: operationalised as age in total months.

IQ: operationalised as the score on a 40-item multiple-choice test.

Aggression: operationalised as the number of aggressive responses in an observed 30 minute period.

Population and sample
Identify the population and sample

IV
State independent variable and how it will be measured

DV
State dependent variable and how it will be measured

Compare to control
Compare to control group

As measured by
How can you tell there has been a result?

Operational Hypothesis

♦ An operationalised hypothesis requires three parts:

1. Sample and Population (that the sample is taken from).
2. Operationalised independent variable.
3. Operationalised dependent variable.

♦ Start it with: It is hypothesised that... It is a prediction so you are writing what you think will happen in the future.

♦ It should only be one sentence long.

Operational hypotheses are detailed predictions for what will occur in an experiment.

Reliability and Validity

► **Reliability:** the extent to which a measurement tool gives consistent results.

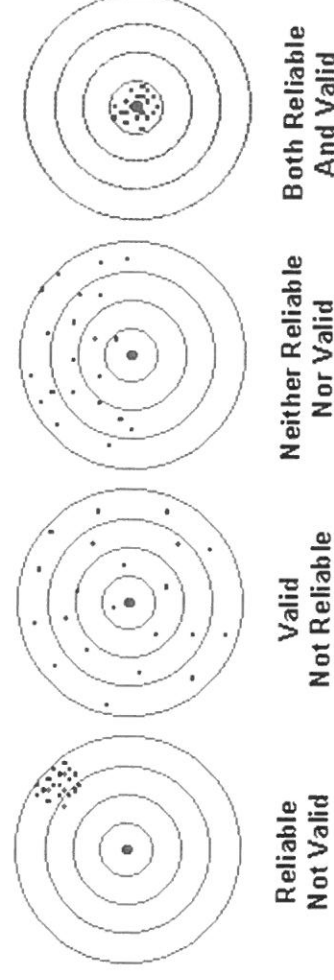
For example; a Likert test – the first half of the questions will repeat in a similar format in the second half

► **Validity:** the extent to which a measurement tool measures what it is supposed to measure is called validity.

For example an IQ test must actually assess general knowledge and abilities, not just mathematical understanding.

Relationship between reliability and validity

- There is actually a relationship between reliability and validity. There are three possible outcomes:
1. **We are far away from the target, but all of our darts land close to each other.** This is an example of when we might have high reliability but low validity.
 2. **Our darts are spread out across the whole board.** In this case, we are neither reliable nor valid
 3. **Our darts are clustered really close together near the centre of the target.** This is the ideal situation; we have a high reliability and high validity.



- ▶ Questionnaires are a series of written questions a participant answers.
- ▶ This most often takes the form of pre-set answers, like a multiple-choice test
- ▶ Questionnaires are usually self-explanatory so that when you hand it to someone, they are able to read and understand what they are supposed to do.
- ▶ Can collect large amounts of data quickly
- ▶ One risk with questionnaires is that they tend to be very limited in what they can ask because you can't expect someone to sit there for two hours to fill out some questions for you.

Interviews

- ▶ Interviews are questions posed to an individual to obtain information about the individual in a structured, semi-structured, or unstructured format.

Structured interview

- ▶ Has planned questions that are asked in a set order. This ensures that everyone is asked the same questions and the same information is collected from all participants.
- ▶ Flaw: does not adapt to the individual being resourced

Unstructured interview

- ▶ Has no predetermined questions and allows the researcher to ask what questions he or she deems important.
- ▶ Provides a researcher with the greatest freedom to get at what they want, but comes at the risk of missing something.

Field Research

- ▶ Research done in the real world
- ▶ Advantage: field research offers psychologist the best view of what happens in real world situations.
- ▶ Disadvantage: uncontrollable

Observational research

- ▶ Involves watching how people act
- ▶ You can remember it because the researcher is observing the participants.

Strongly Disagree	Disagree	Slightly Disagree	Slightly Agree	Agree	Strongly Agree
1	2	3	4	5	6
50% Negative			50% Positive		

NON- EXPERIMENTAL Research Methods

DATA COLLECTION

Surveys

- ▶ Surveys are defined as brief interviews and discussions with individuals about a specific topic.
- ▶ One advantage is that a survey is able to collect a large amount of information relatively quickly because you don't need in-depth background about a person to collect their opinion.
- ▶ A risk of collecting data in this way is most surveys require a brief interview, which means the style and the specific questions asked can vary depending on which research assistant is conducting the interview.
- ▶ Many social topics can be researched using a survey.

Questionnaires

- ▶ Observational research almost always happens in the field, not in a lab.
- ▶ Advantage: a good way to get into the head of the participants
- ▶ Disadvantage: they might not be completely honest.



Method	Description	Benefits	Limitations	How to classify it
Case studies (e.g. Phineas Gage case study)	An in-depth focus on all aspects of a single person. Case is studied in context (e.g. children school) & uses variety of data collection techniques (include observation and interview).	Provide detailed info. Provides insight for future research.	They are uncontrolled studies. Lack formal control groups limiting the conclusions that can be drawn from clinical observations.	Non-experimental Could be either/both quantitative or qualitative depending on data collection techniques used.
Survey (self-rating scale) Usually takes form of a questionnaire. E.g. checklist or Likert scale.	Researcher gathers self-reported data from participants who, ideally, have been randomly selected.	Can analyse data statistically. Can collect data from large Number of people. Can be conducted remotely to reduce/prevent geographical dependence. Cost efficient to study large	Responses limited to options provided. No opportunity for rich data or reasons for responses. Wording effect (phrasing & order of questions can greatly affect people's	Non-experimental Could be either/both quantitative (subjective) or qualitative depending on data collection techniques used

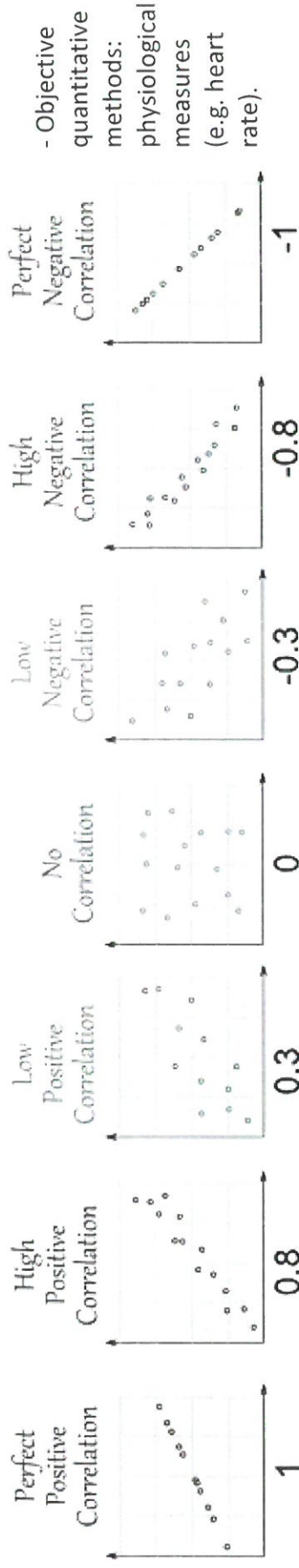
Checklists	Provides Yes/No information on certain statements.	group. Time efficient to study large group. Often administered to different groups of people allowing comparisons to be made across groups.	answers. Responses must be gathered from a representative sample.	Non-experimental Quantitative (Subjective quantitative)
Rating scales (e.g. Likert scales)	Allows people to indicate strength of agreement/ how much they agree with a certain statement.			Non-experimental Quantitative (Subjective quantitative)
Correlational Studies	Study designed to measure the degree of relationship between two existing traits, behaviours or events.	They raise alternative hypotheses that can be then tested in research using an experimental design that can allow cause to be examined.	Finding a correlation between two measures doesn't demonstrate causation.	Non-experimental Could be either/both quantitative or qualitative depending on whether description or numerical data used.
Method	Description	Benefits	Limitations	How to classify it
Archival Research (e.g. using private documents like diaries, public documents held by hospitals).	Examination of old, usually written material found in places like libraries, government buildings, law courts and universities.	Usually cheaper than collecting new data. Allows examination of data gathered over long period of time without taking researcher's lifetime. Allows access to very large, representative samples.	Time needed to identify appropriate data sets & get permission to use the data. Effort required to understand all factors involved with the data. Knowledge and skill required to analyse large and complex data sets.	Non-experimental Quantitative
Physiological measures (e.g. heart rate)	Measurement of physiological structures & functions. A meeting between a participant & an interviewer to collect data for research.	Scientifically recorded data sets and comparable	Data is an average measure set time period	Non-experimental Quantitative (Objective quantitative)
Interview		Rich data collected. Opportunity to discover detail or reasons for responses.	Responses may not be Honest. Huge time required to study group.	Non-experimental Qualitative (Subjective qualitative)

Observational studies/ observations	Observe in a small or large setting	Collection of a large quantity of data	Can be limited by experimenter bias	Non-experimental Qualitative
Independent measures	Random allocation is used to make sure each participant has an equal chance of being assigned to one group or the other.	Increased reliability and validity		Experimental

- ▶
- ▶ A **correlation coefficient** is defined as a numerical representation of the strength and direction of the relationship. It is usually represented by a lowercase 'r.' The correlation coefficient is a number that represents how similar the two variables are. It ranges from -1 to 1.
- ▶
- ▶ Coefficients range from -1 to 1, which describes how scattered the data is. The more like a line it is, the closer it is to 1 (or -1).
- ▶ **Perfect correlation**, defined as a perfect relationship between two variables.
- ▶ Negative implies an inverse correlation, or that when one variable goes up, the other variable goes down.
- ▶ A 0 means there is no correlation.

Correlation

- ▶ A **correlation** is defined as a relationship between two variables. For example, as ice cream sales increase, so do murder rates.
- ▶ Remember: **correlation is not the same as causation**. Two factors may be related without one *causing* the other to occur. Often, a third factor explains the correlation.
- ▶ A **scatterplot** is a visual representation of the relationship between two variables.
 - ▶ places one variable on axis Y and the other variable on axis X



Qualitative and Quantitative methods

UNDERSTANDING DATA

Quantitative measurement is measurement of data that can be put into numbers. The goal of quantitative measurement is to run statistical analysis, so data has to be in numerical form.

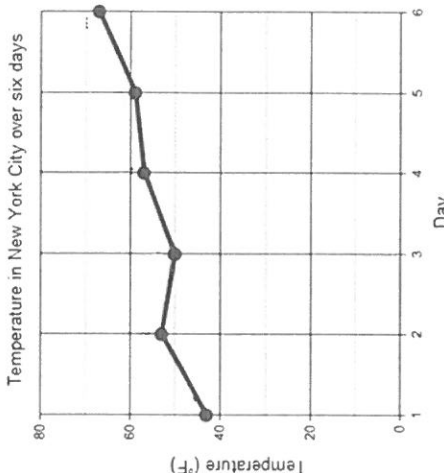
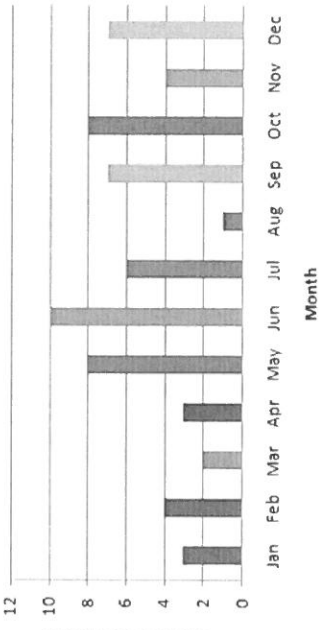
Qualitative measurement focuses on collecting information that is not numerical

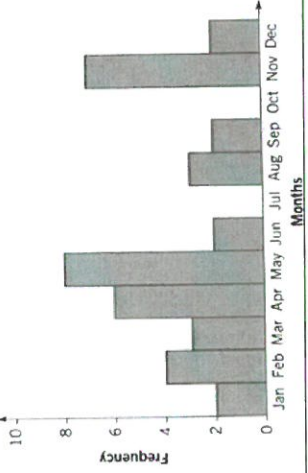
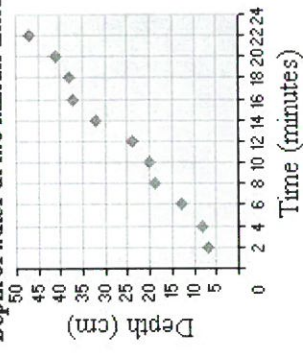
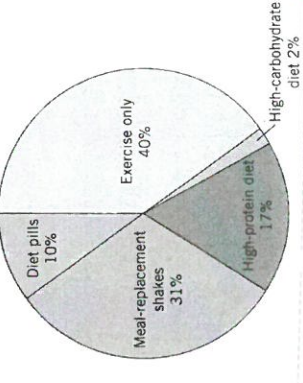
Quantitative methods include: rating scales, Likert scales, checklists, surveys (self-rating scale), questionnaires.

- Subjective quantitative methods: rating scales, Likert scales, checklists, questionnaires.

Types of Graphs

Type of graph	Description	Example
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Line graph	Both variables are increasing/decreasing at regular increments. There is a pattern or trend occurring. Data is continuous for both variables.	Temperature in New York City over six days  <table><tr><th>Day</th><th>Temperature (°F)</th></tr><tr><td>1</td><td>45</td></tr><tr><td>2</td><td>55</td></tr><tr><td>3</td><td>50</td></tr><tr><td>4</td><td>55</td></tr><tr><td>5</td><td>60</td></tr><tr><td>6</td><td>65</td></tr></table>	Day	Temperature (°F)	1	45	2	55	3	50	4	55	5	60	6	65												
Day	Temperature (°F)																											
1	45																											
2	55																											
3	50																											
4	55																											
5	60																											
6	65																											
Bar graph	Bars do not touch. Data is discrete.	Birthday of Students by Month  <table><tr><th>Month</th><th>Number of Students</th></tr><tr><td>Jan</td><td>3</td></tr><tr><td>Feb</td><td>1</td></tr><tr><td>Mar</td><td>2</td></tr><tr><td>Apr</td><td>3</td></tr><tr><td>May</td><td>4</td></tr><tr><td>Jun</td><td>5</td></tr><tr><td>Jul</td><td>6</td></tr><tr><td>Aug</td><td>7</td></tr><tr><td>Sep</td><td>8</td></tr><tr><td>Oct</td><td>9</td></tr><tr><td>Nov</td><td>10</td></tr><tr><td>Dec</td><td>11</td></tr></table>	Month	Number of Students	Jan	3	Feb	1	Mar	2	Apr	3	May	4	Jun	5	Jul	6	Aug	7	Sep	8	Oct	9	Nov	10	Dec	11
Month	Number of Students																											
Jan	3																											
Feb	1																											
Mar	2																											
Apr	3																											
May	4																											
Jun	5																											
Jul	6																											
Aug	7																											
Sep	8																											
Oct	9																											
Nov	10																											
Dec	11																											

Histogram	Class intervals are on the horizontal axis and the frequency is on the vertical axis. The bars are touching. Data is continuous.	Frequency of rainfall over time 
Scatterplot (also called scattergram or scattergraph)	Independent variable on x axis and dependent variable on y axis. OR If there is no dependent variable (two independent variables instead) then can plot variables on either axis. Will show degree of correlation between two variables.	Depth of water at two-minute intervals 
Pie chart	Circular chart divided into sectors. Illustrate relative frequencies or percentages.	Comparison of weight-loss methods used 

Measures of Central Tendency

- ▶ A **measure of central tendency** is a single value that describes the way in which a group of data cluster around a central value.
- ▶ There are three measures of central tendency:
 1. Mean
 2. Median
 3. Mode
- ▶ **Central tendency** is very useful in psychology. It lets us know what is normal or 'average' for a set of data.
- ▶ It also condenses the data set down to one representative value, which is useful when you are working with large amounts of data.
- ▶ **Mean: average**
 - ▶ The **mean** is the preferred measure of central tendency because it considers all of the values in the data set.
 - ▶ Limitations: in order to calculate the mean, data must be numerical.
- 1. Add the data together to find the sum
- 2. Take the sum of the data and divide it by the total number of data
- ▶ **Median**
 - ▶ is the value that cuts the data set in half
- 1. List the data in numerical order
- 2. Locate the value in the middle of the list
- 3. When working with an even number of data, you must find the mean of the two middle numbers
- ▶ **Mode**
 - ▶ The mode is the only measure of central tendency that can be used with both nominal and numerical data.
 - ▶ The mode is defined as the value that appears most often.
 - ▶ If every value only appears once, there is no mode.

- ▶ There can be more than one mode!

- ♦ **Mean:** a measure of central tendency found by adding up the values in the data set then dividing by the number of values that were added.

Example :

Data Set = 2, 5, 9, 3, 5, 4, 7

Number of Elements in Data Set = 7

Mean = $(2 + 5 + 9 + 7 + 5 + 4 + 3) / 7 = 5$

- ♦ **Median:** a measure of central tendency found by listing the values of the data set in numerical order and identifying the value that appears in the middle of the list.

The "Median" of a data set is dependant on whether the number of elements in the data set is odd or even. First reorder the data set from the smallest to the largest then if the number of elements are odd, then the Median is the element in the middle of the data set. If the number of elements are even, then the Median is the average of the two middle terms.

Examples : Odd Number of Elements

Data Set = 2, 5, 9, 3, 5, 4, 7

Reordered = 2, 3, 4, 5, 5, 7, 9

Median = 5

Examples : Even Number of Elements

Data Set = 2, 5, 9, 3, 5, 4

Reordered = 2, 3, 4, 5, 5, 9

Median = $(4 + 5) / 2 = 4.5$

- ♦ Two advantages of using the median instead of the mean as a measure of central tendency is that:
 - The median is not affected by extreme

scores/outliers. - The median is not affected by/does not depend on the size/value of the other scores.

Role of Probability

♦ Probability refers to the likelihood of an event occurring. Tests of statistical significance provide an estimate of how often experimental results might have occurred by chance alone. The results of a significance test are stated as a probability, or *p*-value. This probability gives the odds that an observed difference is due to chance alone. In psychology, an experimental result that could have occurred by chance is five times (or fewer) out of 100 (in other words, less than 5% probability, or $p < 0.05$) is considered statistically significant.

♦ **Statistical significance:** It is a statistic that provides an estimate of how often experimental results could have occurred by chance alone. It is expressed as '*p*-value.'

Statistical significance		
Summary of <i>p</i> -values		
IF <i>p</i> IS LESS THAN 0.05 ($p < 0.05$)	IF <i>p</i> IS GREATER THAN 0.05 ($p > 0.05$)	
The difference between control and experimental groups is statistically significant	The difference between control and experimental groups is <i>not</i> statistically significant	
The difference is <i>unlikely</i> to be due to chance alone	The difference between the groups is <i>likely</i> to be due to chance alone	nificant
The difference between the groups is <i>likely</i> to be due to the IV	The difference is <i>unlikely</i> to be due to the IV	ly,
The experimental hypothesis is <i>supported</i>	The experimental hypothesis is <i>rejected</i>	how often alone.

♦ If you conduct a study with a significance level set at $p < 0.05$, then the following will be true:

- If p is less than or equal to 0.05 (for example, 0.04), then the difference between the experimental group's results and the control group's results is said to be statistically significant (that is, not due to chance alone), and likely due to the effect of the IV (independent variable). The experimental hypothesis is therefore supported.

- If p is greater than 0.05 (for example, 0.08), then the difference between the experimental group's results and the control group's results is said not to be statistically significant (that is, it is likely due to chance alone). The experimental hypothesis is rejected.

Sources of error in data and ways of reducing these

♦ There are potential sources of error in data that emanate from the participants and from the experimenter.

Sources of error in data could include: participants not randomly selected, participants in groups not randomly allocated, experimenter effect/experimental bias, participants changing behaviour due to predictions regarding research.

1. Random sampling can be used to ensure that the sample is representative of the population.

2. Random allocation is used to allocate participants to groups and can be used as a way to reduce differences between experimental and control groups. (Random allocation is a concept of 'independent measures', an example of a type of experimental research method).

3. Try to eliminate the experimenter effect.

4. Use a placebo to reduce participant error (when participants change their behaviour due to being part of a study).

Bias

- ▶ Scientific experiments are usually structured with a control group and an experimental group.
- ▶ The **control group** is the neutral group - they do not receive the treatment or have any other experimental condition applied to them.
- ▶ The **experimental group**, or treatment group, receives the treatment or other variable. A control group is necessary in order to test the effects of the experimental variable on the experimental group.

Practical issues associated with planning and conducting research include:

- Obtaining ethical consent before being able to carry out the research.
- Making sure the ethical guidelines are followed throughout the research.
- Obtaining an appropriate sample.
- Carrying out research that will be valid as well as reliable.
- Carrying out research that will be statistically significant.
- Trying to eliminate uncontrolled variables.
- Trying to eliminate the experimenter effect.

Extraneous Variables

- ▶ **Extraneous variables**, or variables present in the experiment that aren't being studied.

- ▶ The problem with extraneous variables is that they might affect the dependent variable but they might not.
- ▶ There's no way to tell until after the experiment is done.
- ▶ Extraneous variables are usually grouped into three categories:
 1. **Physical or Situational Variables:** These occur when the physical situation of subjects changes for certain groups, like a warmer room.
 2. **Personal Variables:** These are when one group has personality or other traits that members of the other group don't.
 3. **Researcher Variables:** These are when the researcher, himself, does something different for the various groups of the experiment. For example, if the experiments is warmer towards one group.

Confounding Variables

- ▶ A confounding variable is an extraneous variable that is related to your independent variable and might affect your dependent variable.
- ▶ In an ideal study, there will be no confounding variables.

Evidence-based conclusions related to the hypothesis

- ♦ When psychologists carry out experiments, they usually want to do more than describe and summarise data. They want to be able to determine whether their hypotheses have been supported by the data. To do this, they use inferential statistics, statistics that let them draw inferences and make conclusions from the data.
- ♦ There are many different types of inferential statistics that are beyond the scope of your course. The specific type used depends on the type of data being collected, the type of study being done, and what the

researcher wants to know about the results. Researchers typically want to draw conclusions about the larger population from which their sample of participants was drawn.

- ♦ The researchers' ability to come to conclusions about the population will depend on all the factors we have considered:
 - Selection of a representative sample.
 - Appropriate allocation of participants to experimental and control groups.
 - Ensuring participants are unaware of which group they have been allocated to.
 - Eliminating the possibility of experimenter bias.

FLASHCARDS

Use these to recall the concept.

Phase two of these is demonstrating